Waldenström macroglobulinemia

Lymphoplasmacytic lymphoma with an IgM paraprotein · **MYD88 L265P in ~90%+** · treat for symptoms and IgM complications, not IgM level · genotype (MYD88/CXCR4) steers BTKi vs chemoimmunotherapy

Compiled from primary trial reports and consensus guidelines (NCCN; IWWM-2, IWWM-11 and IWWM-12 consensus panels) verified against PubMed, September 2026.

DEFINITIONS AND IGM DIFFERENTIAL

Entity	Defining features
WM	**Lymphoplasmacytic lymphoma (LPL) in the marrow + an IgM monoclonal protein of any size.** IWWM-2 and IWWM-11: **no IgM or marrow-% cutoff** separates WM from IgM MGUS — the diagnosis rests on marrow histology + immunophenotype
IgM MGUS	IgM M-protein, no LPL on marrow, no symptoms. Progresses to lymphoma/WM at **~1.5%/yr**. IWWM-11 splits it: **plasma-cell type** (MYD88 wild-type — myeloma/AL lineage) vs **B-cell type** (may be MYD88-mutant)
Asymptomatic (smoldering) WM	Meets WM criteria, no treatment indication → **observe**. **72%** progress (median follow-up 7.8 yr). Progression risk: **IgM ≥4,500 mg/dL, marrow ≥70%, β2M ≥4 mg/L, albumin ≤3.5**; MYD88 wild-type HR 2.7
IgM myeloma	**CD138+ plasma cells without a B-cell/LPL component, t(11;14), MYD88 wild-type, lytic lesions.** Treat as myeloma — not WM
Marginal zone	Can make IgM and show plasmacytic differentiation. **MYD88 L265P absent or rare** — the most useful discriminator
IgM-related disorder	Symptoms from the IgM itself (neuropathy, cryo, CAD, amyloid) **without overt lymphoma** — still may need clone-directed therapy

WORKUP

- **SPEP + immunofixation, quantitative IgM** (SPEP M-spike and nephelometric IgM are not interchangeable — follow one method), free light chains, β2-microglobulin, albumin, LDH, CBC, CMP
- **Marrow biopsy**: intertrabecular LPL infiltrate; surface IgM+, CD19+, CD20+, CD22+, with CD138+ plasma cells; usually CD5– CD10– CD23– (variants do not exclude WM)
- **Molecular on marrow before any therapy** (IWWM-11): **AS-PCR for MYD88 L265P and CXCR4 S338X on whole marrow**; CXCR4 + TP53 sequencing and 6q/17p FISH on CD19-selected cells
- **CT chest/abdomen/pelvis** for nodes/spleen. **Serum viscosity + fundoscopy if IgM is high or any HVS symptom**
- Targeted: **cryoglobulins** (draw and transport at 37°C), **DAT + cold agglutinin titer**, **anti-MAG** + EMG if neuropathy, **NT-proBNP/troponin, urine protein, fat pad** if amyloid suspected, **MRI brain/spine + CSF** if CNS symptoms
- **Prognosis:** IPSSWM — age >65, Hb ≤11.5, plt ≤100, β2M >3 mg/L, M-protein >7 g/dL; 5-yr OS **87%** / **68%** / **36%** (low/int/high). **Not a treatment trigger**

TREATMENT PATHWAY

Diagnosis

Asymptomatic → watch and wait
IWWM: **IgM level alone is NOT an indication**
clinic + labs q3–6 mo; IgM MGUS yearly SPEP

Treat when a criterion is met
Hb ≤10 or plt <100 from marrow infiltration
B symptoms, bulky/symptomatic nodes or spleen
HVS, neuropathy, amyloid, cryo, CAD, renal

Emergencies

Symptomatic hyperviscosity
plasmapheresis FIRST, then start therapy
no RBC transfusion before pheresis

High IgM + rituximab planned
hold rituximab until IgM falls, or pheresis first — the IgM flare can precipitate HVS

First line — genotype + patient

Fixed-duration chemoimmunotherapy
BR ×6 or **DRC ×6** — IWWM default, affordable
MYD88-WT, CXCR4-mutant, fit, want time off

Bortezomib-based (BDR / B-DRC)
rapid IgM control: HVS, cryo, amyloid, renal
avoid with pre-existing neuropathy

Continuous covalent BTKi
zanubrutinib preferred (ASPEN); ibrutinib ± R
MYD88-mutant, CXCR4-WT, frail, CNS disease
not for AL amyloid (IWWM-11)

Relapse

Switch class; reuse only after a long remission
after chemo → cBTKi; after cBTKi → BR/DRC/BDR
pirtobrutinib or **venetoclax** (off-label)
bridge — don't stop a BTKi until next Rx

IGM-MEDIATED COMPLICATIONS

Syndrome	Recognize → manage
Hyperviscosity *large pentameric IgM*	Blurred vision, headache, epistaxis/oronasal bleeding, dizziness, confusion, heart failure. **Fundoscopy:** dilated "sausage" veins, hemorrhages, papilledema. Viscosity poorly predicts symptoms per patient. **Plasmapheresis first** (IgM is ~80% intravascular, so it clears efficiently), **then cytoreduce** — pheresis alone is a bridge. **Avoid RBC transfusion until pheresed**
Anti-MAG neuropathy	Slowly progressive, distal, **symmetric sensory-predominant demyelinating** neuropathy with ataxia/tremor. EMG + anti-MAG titer; joint heme-neuro care. **Rituximab-based therapy**; stabilization is the realistic goal. **Avoid bortezomib** (and vincristine)
Cryoglobulinemia *type I or II*	Raynaud's, acral necrosis, purpura, vasculitic neuropathy, GN. **Draw at 37°C** or it is missed. Keep warm; **plasmapheresis for severe**; rapid cytoreduction (BR, BDR)
Cold agglutinin disease	Chronic hemolysis with cold acrocyanosis; DAT C3d+, IgMκ titer. **Keep warm, warm IV fluids/blood**. Rituximab-based (BR most active); **sutimlimab** blocks C1s for hemolysis
AL amyloidosis	Lambda light chain, cardiac/renal/nerve. **Treat as AL**: bortezomib- or bendamustine-based; daratumumab. **cBTKi not recommended** (IWWM-11)
Bing-Neel *CNS WM*	Headache, cognitive change, cranial nerve palsies, gait. **MRI brain + spine with gadolinium, CSF cytology/flow + MYD88 PCR. BTKi cross the BBB**: ibrutinib (n=28: 85% symptom improvement, 2-yr EFS 80%); IWWM-12 made **zanubrutinib a standard**. Alternatives: HD-MTX / HD-araC, fludarabine

GENOTYPE AND BTKI CHOICE

Lesion	Why it matters
MYD88 L265P *~90%+*	Drives NF-κB via IRAK/BTK. **Supports the diagnosis** (rare in MZL, absent in myeloma). **MYD88 wild-type:** weaker BTKi response (ibrutinib major response 29%), worse OS — favor chemoimmunotherapy
CXCR4 WHIM-like *~30–40%*	Nonsense/frameshift C-tail mutations; higher IgM + marrow burden, more hyperviscosity. **Slower, shallower BTKi responses**: TN ibrutinib major response 71% vs 94%, VGPR 7% vs 31%, time to major response 7.3 vs 1.8 mo
TP53, del6q, del17p	Adverse; IWWM-11 counts TP53 and 6q/17p among essential pre-treatment tests

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FIRST-LINE OPTIONS — EVIDENCE

Regimen	Evidence and the catch
Bendamustine-rituximab	IWWM/NCCN preferred chemoimmunotherapy. Randomized evidence is **StiL NHL1** (BR beat R-CHOP in indolent NHL, WM a small subgroup). **Rituximab maintenance after BR: no benefit** overall (StiL NHL7-MAINTAIN)
DRC	Phase 2, n=72: ORR **83%** (CR 7%), 2-yr PFS 67%; grade 3–4 neutropenia only 9%. Slow: median time to response **4.1 mo**. **ECWM-1: adding bortezomib (B-DRC) did not improve PFS** (24-mo 80.6% vs 72.8%, P=0.32) but responses came faster (3.0 vs 5.5 mo)
BDR	Weekly-bortezomib EMN regimen: ORR **85%**, median PFS **42 mo**; neuropathy **46%** (grade ≥3 7%). Bortezomib lead-in blunts rituximab flare. **Zoster prophylaxis mandatory**
Ibrutinib ± rituximab	**iNNOVATE** (phase 3, n=150): IR vs placebo-R — final median PFS **NR vs 20.3 mo**, HR 0.25; benefit **regardless of MYD88/CXCR4**. Adding ibrutinib cut rituximab IgM flare (8% vs 47%). Grade ≥3 **AF 12% vs 1%**, HTN 13% vs 4%
Zanubrutinib	**ASPEN** (phase 3, n=201 MYD88-mutant): primary endpoint VGPR+CR **28% vs 19%, P=0.09 — not met**. Final (44 mo): VGPR+CR 36% vs 25%; CXCR4-mutant 21% vs 10%. **Less AF (7.9% vs 23.5%), HTN, bleeding, diarrhea**; more neutropenia. FDA-approved Aug 2021
Acalabrutinib	Single-arm phase 2, n=106: ORR **93%** TN and R/R; grade 3–4 AF 1%. **Not FDA-approved for WM**
Ibrutinib + venetoclax	Fixed-duration TN trial: VGPR 42% but **ventricular arrhythmia 9% incl. 2 deaths → stopped early**. Not a standard

FIXED-DURATION VS CONTINUOUS

- **Chemoimmunotherapy:** finite (~6 mo), then time off therapy; myelosuppression, infections; the **better fit for MYD88 wild-type** and a reasonable one for CXCR4-mutant disease
- **cBTKi:** oral, rapid Hb recovery, **indefinite** — cost, cumulative AF/HTN/bleeding, and **IgM rebound if stopped**. Best in MYD88-mutant/CXCR4-WT; zanubrutinib over ibrutinib for tolerability
- **IgM lags the marrow:** IgM can plateau or rise transiently (rituximab flare; **bortezomib/rituximab-era discordance**) — do not call progression on IgM alone; confirm, check marrow/Hb/nodes. BR/DRC responses deepen for months after the last cycle

RELAPSED / REFRACTORY

- **Durable first remission → the same regimen can be reused**; early relapse or refractory → switch class
- **Ibrutinib withdrawal:** IgM rebound (≥25%) in **73%** of discontinuations, half within 4 weeks, some with HVS; salvage works better if started within 2 weeks. **Continue the BTKi until the next therapy starts**; hold only ~3–7 days peri-procedure
- **Pirtobrutinib** (non-covalent; binds C481-mutant BTK): BRUIN WM n=80 (79% cBTKi-exposed) — ORR **82.5%** (81% post-cBTKi), major response ~72%. **Not FDA-approved for WM** (MCL/CLL only) — single-arm, off-label
- **Venetoclax** 200→800 mg daily ×2 yr: ORR 84%, major 81%, median PFS **30 mo**; slower after BTKi (4.5 vs 1.4 mo); **CXCR4 did not matter**. Neutropenia 45%. Off-label
- Also: bortezomib/carfilzomib-based, BR after BTKi, clinical trials (BTK degraders, CXCR4 antagonists, bispecifics). **Autologous SCT** for select chemosensitive young patients. Transformation to DLBCL → treat as DLBCL

RESPONSE (IWWM-6 / SIMPLIFIED IWWM-11)

CR	Normal IgM, **negative immunofixation**, no marrow disease, resolved nodes/spleen
VGPR	≥90% IgM reduction
PR	50–89% IgM reduction · **major response = PR or better**
MR	25–49% IgM reduction (counts in "overall" response)
PD	≥25% IgM rise from nadir (confirmed) or new disease/complication

IWWM-11 lets PR/VGPR be called on serum IgM alone, with guidance for flare/rebound. **ORR in WM trials includes minor responses** — compare major response rates across studies.

REGIMEN DOSING

Regimen	Dosing
BR	Bendamustine **90 mg/m² d1–2** + rituximab **375 mg/m²** d1, **q28d ×6**
DRC	Dexamethasone **20 mg IV** + rituximab **375 mg/m²** d1 + cyclophosphamide **100 mg/m² PO BID d1–5**, q21d ×6
BDR *EMN*	Cycle 1 bortezomib alone 1.3 mg/m² d1,4,8,11; then **1.6 mg/m² weekly d1,8,15,22 q35d ×4**; dex 40 mg + rituximab 375 in cycles 2 and 5. SC bortezomib in practice
Zanubrutinib	**160 mg BID or 320 mg daily**, continuous
Ibrutinib ± R	**420 mg daily**, continuous; iNNOVATE rituximab 375 mg/m² weekly, **weeks 1–4 and 17–20**
Acalabrutinib	100 mg BID, continuous
Pirtobrutinib	200 mg daily, continuous
Venetoclax	Ramp 200 → **800 mg daily**, up to 2 yr; TLS precautions

MONITORING AND PROPHYLAXIS

- **HBsAg + anti-HBc before any anti-CD20** — entecavir if core-Ab positive. **Acyclovir/valacyclovir with bortezomib** (zoster in 4 of the first 7 BDR patients) and commonly with BR and BTKi
- **BTKi:** baseline ECG and BP; ask about palpitations; **treat HTN**. AF → rate control, weigh anticoagulation with bleeding; switch ibrutinib → zanubrutinib. **No warfarin; avoid fish oil/vit E**; hold ~3–7 days around surgery. Strong CYP3A inhibitors → dose-adjust
- **Bortezomib:** neuropathy check every cycle; **weekly + SC** dosing lowers it; dose-reduce at grade 2 with pain
- **Rituximab:** IgM before each dose; **hold or pre-pherese if IgM is high** (commonly ≥4,000 mg/dL); a rising IgM in the first months is often flare, not progression
- IgM + CBC q1–3 mo on therapy; SPEP/IgM q3–6 mo in remission. **Same lab, same method**

KEY TRIALS

iNNOVATE	Ibrutinib-R vs placebo-R — median PFS NR vs 20.3 mo, HR 0.25, genotype-independent
ASPEN	Zanubrutinib vs ibrutinib — VGPR+CR primary **not met** (28 vs 19%); far less AF
ECWM-1	B-DRC vs DRC — no PFS gain; faster response
DRC (Dimopoulos)	Phase 2 — ORR 83%, 2-yr PFS 67%
BDR (EMN)	Phase 2 — ORR 85%, median PFS 42 mo, weekly bortezomib
Ibrutinib TN / R/R	Phase 2 — CXCR4 and MYD88 genotype set depth and speed
BRUIN WM	Pirtobrutinib R/R — ORR 82.5%, active post-cBTKi
Venetoclax (Castillo)	R/R — major response 81%, median PFS 30 mo